1. Affine Hecke algebras

In this section, we will discuss representation theory of affine Hecke algebra of classical types — the origin of quiver with involution.

1.1. Affine Hecke algebra as convolution algebra. By [KL87], as an algebra

$$\text{Affine Hecke algebra} \cong K^{\mathbb{G}}(\mathcal{Z}) \qquad \mathcal{Z} = \tilde{\mathcal{N}} \times_{\mathcal{N}} \tilde{\mathcal{N}};$$

see also [CG, Chapter 7]. Here

- $\mathcal{N} = \{x \in \mathfrak{g} : x \text{ is nilpotent}\}$;
- $\tilde{\mathcal{N}} = T^*\mathcal{B} = \{(x, B) \in \tilde{\mathcal{N}} \times \mathcal{B} : x \in B\} \cong G \times_B \mathfrak{n}$.
- $\pi : \tilde{\mathcal{N}} \to \mathcal{N}$ is the natural projection.
- $\mathbb{G} = G \times \mathbb{C}^{\times}_{\text{dia}}$.

Remarks:

- Actually, the affine Hecke algebra is actually for the Langlands dual group G^L.
- The element T_i corresponds to certain Hecke correspondence; and the element X^λ corresponds to the correspondence of tensor product with a line bundle (in particular, it is supported on the diagonal).

1.2. Fixed points reduction. There is a standard way to study the representation theory of a convolution algebra; see [CG]. Let me briefly discuss. Assume we are given a semisimple element $(s, q) \in \mathbb{G}$.

- The center of the LRS is

$$\mathbb{C}[\text{character lattice}]^W[\mathbf{q}^{\pm 1}] = R(G)[\mathbf{q}^{\pm 1}].$$

  Then (s, q) defines a center character. Explicitly, by a conjugation, we can assume $s \in T$, and it makes sense to substitute into characters; and $\mathbf{q} \mapsto q$.
- We can take fixed points loci

$$\mathcal{Z}^{(s,q)} = \tilde{\mathcal{N}}^{(s,q)} \underset{\mathcal{N}^{(s,q)}}{\times} \tilde{\mathcal{N}}^{(s,q)}.$$

  We have an isomorphism of algebra

$$\begin{matrix}\text{affine Hecke algebra} \\ \text{specialized by } (s,q)\end{matrix} \cong H_*(\mathcal{Z}^{(s,q)}).$$

- The problem of classifying simple modules finally reduces to study the perverse sheaves on $\mathcal{N}^{(s,q)}$. In this step, it is useful to add back the equivariance of $\mathbb{G}^{(s,q)} = Z_G(s) \times \mathbb{C}^{\times}$. In summary, the simple modules are parametrzied by

$$(s, q) \qquad \& \qquad \text{some local systems on strata.}$$

1.3. **Type A case.** Let us work with GL_n. We have the explicitly identification:

- $\mathcal{N} = \{\text{nilpotent matrices in } \mathfrak{gl}_n\}$;
- $\mathcal{B} = Fl_n = \{0 = V_0 \subset V_1 \subset \cdots \subset V_{n-1} \subset V_n = \mathbb{C}^n : \dim V_i = i\}$;
- $\tilde{\mathcal{N}} = \{(A, V_\bullet) \in \mathcal{N} \times Fl_n : AV_i \subseteq V_{i-1}\}$.

Let us choose (s, q) with $q \neq 1$. By conjugation, we can assume s is diagonal, but this is not necessary. Then

$$\mathcal{N}^{(s,q)} = \{A \in \mathcal{N} : sAs^{-1} = qA\} = \{A \in \mathcal{N} : sA = qAs\}.$$

Let us decompose

$$\mathbb{C}^n = \bigoplus_{\lambda \in \mathrm{Spec}(s)} V_\lambda, \qquad \begin{array}{c} \mathrm{Spec}(s) = \{\text{eigenvalues of } s\}, \\ V_\lambda = \text{eigenspace of } \lambda = \{x \in \mathbb{C}^n : sx = \lambda x\}. \end{array}$$

We have

$$\mathcal{N}^{(s,q)} = \bigoplus_{\lambda, q\lambda \in \mathrm{Spec}(s)} \mathrm{Hom}(V_\lambda, V_{q\lambda}).$$

Here is the diagram explaining this equality:

$$\begin{tikzcd} V_\lambda \arrow[loop above, "\lambda"] \arrow[r, "A"] & V_{q\lambda} \arrow[loop above, "q\lambda"] \end{tikzcd} \qquad \begin{array}{c} \text{the action of } s \\ \text{the action of } A. \end{array}$$

The key observation is

$$\mathcal{N}^{(s,q)} = E_V = \text{Lusztig's quiver variety of } Q$$

where

$$Q = (Q_0, Q_1), \qquad \begin{array}{c} \text{vertex } Q_0 = \mathrm{Spec}(s), \\ \text{arrows } Q_1 : \lambda \to \mu \iff \mu = q\lambda. \end{array}$$

With more efforts, one could identify

$$\tilde{\mathcal{N}}^{(s,q)} = \tilde{\mathcal{F}}_V = \text{Lusztig's quiver flag variety of } Q.$$

Theorem (Ariki [Ari]**).** We have

$$K \begin{pmatrix} \text{the block of rep's of} \\ \text{Affine Hecke algebra} \\ \text{determined by } (s,q) \end{pmatrix} \cong \begin{array}{c} \text{the component of } U_q(\mathfrak{g}_Q)^+ \text{ of} \\ \text{weight } \sum_{\lambda \in \mathrm{Spec}(s)} (\dim V_\lambda) \alpha_\lambda \end{array}$$

with simple modules correspond to dual canonical basis.

Remark. Note that we assume $q \neq 1$ to avoid the "quiver"

$$\tilde{\mathbb{A}}_0 : \qquad \begin{tikzcd} \circ \arrow[loop right] \end{tikzcd}$$

which is not a quiver.

1.4. **Type BCD.** It is natural to expect similar results for classical groups. Recall

- For SO_n, the adjoint representation $\mathfrak{so}_n \cong \Lambda^2 V$ where $V = \mathbb{C}^n$;
- For Sp_{2n}, the adjoint representation $\mathfrak{sp}_{2n} \cong S^2 V$ where $V = \mathbb{C}^{2n}$.

By [Kato], for group Sp_{2n}, it is better to consider the exotic Springer theory. Precisely,

- we replace the adjoint representation $\mathfrak{sp}_{2n} \cong S^2 V$ by $\Lambda^2 V \oplus V$. Compare the weights
$$\begin{array}{lll} \mathfrak{sp}_{2n} & \{\pm\epsilon_i \pm \epsilon_j : i \neq j\} \cup \{\pm 2\epsilon_i : 1 \leq i \leq n\} & \text{type C} \\ \Lambda^2 V \oplus V & \{\pm\epsilon_i \pm \epsilon_j : i \neq j\} \cup \{\pm \epsilon_i : 1 \leq i \leq n\} & \text{type B.} \end{array}$$
- We replace the nilpotent subalgebra $\mathfrak{n}$ by the span of vectors whose weight is a positive root in type B root system.
- We are able to define exotic Springer resolution
$$\tilde{\mathcal{N}}_e := G \times_B \mathfrak{n}_e \longrightarrow \mathcal{N}_e := \text{it image} \subset \Lambda^2 V \oplus V.$$
- We are able to define an action of $\mathbb{G}_e = Sp_{2n} \times \mathbb{C}^\times \times \mathbb{C}^\times$, since now we have two summands to scale.

By [Kato], as an algebra

$$\begin{array}{c}\text{Affine Hecke algebra} \\ \text{with unequal parameters}\end{array} \cong K^{\mathbb{G}}(\mathcal{Z}_e) \qquad \mathcal{Z}_e = \tilde{\mathcal{N}}_e \times_{\mathcal{N}_e} \tilde{\mathcal{N}}_e.$$

Let $(s, p, q) \in \mathbb{G}_e$ be a semisimple element. We will assume

$$\pm 1 \notin \mathrm{Spec}(s), \qquad q \neq \pm 1.$$

By a similar manner, we get

$$\begin{array}{c}\text{Affine Hecke algebra} \\ \text{with unequal parameters} \\ \text{specialized by } (s,p,q)\end{array} \cong K^{\mathbb{G}}(\mathcal{Z}_e^{(s,p,q)}).$$

We have

$$\mathcal{N}_e^{(s,p,q)} = \left\{ (A, v) \in \mathcal{N}_e : \begin{array}{c} sA = qAs \\ sv = pv \end{array} \right\}.$$

Let us decompose $V = \mathbb{C}^{2n}$ according to $\mathrm{Spec}(s)$ as before. We have

$$\mathcal{N}_e^{(s,p,q)} = \mathrm{Alt}\left(\bigoplus_{\lambda, q\lambda \in \mathrm{Spec}(s)} \mathrm{Hom}(V_\lambda, V_{q\lambda}) \right) \oplus V_p.$$

Let us explain the "Alt" here. First, the symplectic form ω on V commutes with s, and thus build a perfect pairing between

$$\omega : V_\lambda \otimes V_{\lambda^{-1}} \longrightarrow \mathbb{C} \qquad \Longrightarrow \qquad V_\lambda^* \cong V_{\lambda^{-1}}.$$

The fact $A \in \Lambda^2 V$ is equivalent to require the following condition

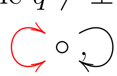
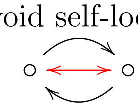
$$\text{if} \qquad V_\lambda \xrightarrow{f} V_{q\lambda} \qquad \text{then} \qquad V_{\lambda^{-1}} \cong V_\lambda^* \xrightarrow{-f^*} V_{q\lambda}^* \cong V_{q^{-1}\lambda^{-1}}.$$

Remarks:

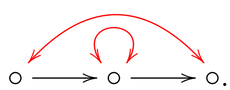
- If we use the classical Springer resolution, we have
$$\mathcal{N}^{(s,q)} = \mathrm{Sym}\left(\bigoplus_{\lambda, q\lambda \in \mathrm{Spec}(s)} \mathrm{Hom}(V_\lambda, V_{q\lambda}) \right).$$
- Similar construction can be done for type D. We have
$$\mathcal{N}^{(s,q)} = \mathrm{Alt}\left(\bigoplus_{\lambda, q\lambda \in \mathrm{Spec}(s)} \mathrm{Hom}(V_\lambda, V_{q\lambda}) \right).$$

Theorem [VV]**.** We have
$$K \begin{pmatrix} \text{the block of rep's of} \\ \text{Affine Hecke algebra} \\ \text{with unequal parameters} \\ \text{determined by } (s,p,q) \end{pmatrix} \cong \begin{matrix} \text{a component of} \\ \text{a module of} \\ \text{EK algebra} \end{matrix}$$
with simple modules correspond to dual canonical basis.

Remark. Note that we assume $q \neq \pm 1$ to avoid self-loop and multi-edges:

 , 

We assume $\pm 1 \notin \mathrm{Spec}(s)$ to avoid fixed points



Actually, the above story about convolution algebra still hold if we have a fixed point. The problem is the identification with the EK-algebra module.

1.5. **Quivers with involution.** A quiver with involution is a pair $(Q, \tau : Q_0 \to Q_0)$ such that

- τ has no fixed point, i.e. $\tau(i) \neq i$, and
- if $i \to j$, then $\tau(i) \leftarrow \tau(j)$.

If we allow multi-edges, then we need to specialize $\tau : Q_1 \to Q_1$ and replace the second condition by

- $h : i \to j$ then $\tau h : \tau j \to \tau i$.

For an Q_0-graded vector space V, recall
$$E_V = \bigoplus_{i \to j} \mathrm{Hom}(V_i, V_j), \qquad G_V = \bigoplus_{i \in Q_0} GL(V_i).$$
Assume further that V is equipped with an pairing such that $V_i^* \cong V_{\tau(i)}$, we can define
$$\begin{aligned} E_V^\tau &= \{(f_{i \to j}) \in E_V : f_{i\to j}^* = -f_{\tau j \to \tau i}\} \\ G_V^\tau &= \{(g_i) \in G_V : g_i^* = g_{\tau i}\}. \end{aligned}$$

Note that this implies

$$\text{the dimension vector } \mathbf{v} = (\dim V_i)_i \text{ is } \tau\text{-invariant.}$$

Assume we have another Q_0-graded vector space W (called framing), we can define

$$L_{W,V} = \bigoplus_{i \in Q_0} \mathrm{Hom}(W_i, V_i), \qquad G_W = \bigoplus_{i \in Q_0} GL(W_i).$$

Returning to the above computation, we can realize

$$\mathcal{N}_e^{(s,p,q)} \cong E_V^\tau \oplus L_{W,V}.$$

where the involution τ on $\mathrm{Spec}(s)$ is given by $\lambda \mapsto \lambda^{-1}$, and

$$V = \bigoplus_{\lambda \in \mathrm{Spec}(\lambda)} V_\lambda, \qquad W = \begin{cases} \mathbb{C} \text{ of degree } p & \text{if } p \in \mathrm{Spec}(s). \\ 0, & \text{otherwise.} \end{cases}$$

Remark. There is a Hall algebra realization of $\imath$quantum groups [LW], but the edge convention is opposite, i.e. if $i \to j$ then $\tau i \to \tau j$.

1.6. **Enomoto–Kashiwara algebra.** Using flag variety of type C, we can construct a resolution $\tilde{F}_V^\tau$ of $E_V \oplus L_{W,V}$, which includes $\tilde{\mathcal{N}}_e^{(s,p,q)}$ as a special case.

Theorem [VV]**.** We have

$$K\big(\text{the convolution algebra}\big) \cong \begin{matrix}\text{the component of a module} \\ \text{of EK algebra of certain weight.}\end{matrix}$$

The weight is determined by $(\dim V_i)_i$, and the module is determined by $(\dim W_i)_i$.

The Enomoto–Kashiwara algebra [EK] is defined for Dynkin diagrams with involution:

Enomoto–Kashiwara's algebra

$$\boxed{\begin{array}{l} \qquad {}^\tau\mathbf{B} = \big\langle T_i^{\pm 1}, E_i, F_i \big\rangle_{i \in I} \\ T_i T_j = T_j T_i, \qquad T_{\tau(i)} = T_i \\ T_i E_j T_i^{-1} = q^{c_{ij} + c_{\tau i, j}} E_j \\ T_i F_j T_i^{-1} = q^{-c_{ij} - c_{\tau i, j}} F_j \\ E_i F_j - q^{-c_{ij}} F_j E_i = \delta_{ij} + \delta_{\tau i, j} T_i \\ \text{Serre relations for } E_i\text{'s and } F_i\text{'s} \end{array}}$$

The generators act as functors on certain category of perverse sheaves over $E_V^\tau \oplus L_{W,V}$. Roughly speaking,

$$E_i = \text{induction},$$
$$F_i = \text{restriction},$$
$$T_i = \text{grading}.$$

Remark. The presentation of ${}^{\tau}\mathbf{B}$ is very different from $\imath$QG (the coideal subalgebra of QG):

$$\begin{array}{c}\imath\text{Quantum groups}\\ \boxed{\begin{array}{l} \qquad\quad \mathbf{U}^{\imath} = \left\langle k_i^{\pm 1}, B_i \right\rangle_{i\in I} \\ k_i k_j = k_j k_i, \qquad k_{\tau i} = k_i^{-1} \\ k_i B_j k_i^{-1} = q^{c_{ij} - c_{\tau i, j}} B_j \\ B_i B_j - B_j B_i \overset{(c_{ij}=0)}{=\!=} \delta_{\tau i, j} \dfrac{k_i - k_i^{-1}}{q - q^{-1}} \\ \imath\text{Serre relations for } B_i\text{'s} \end{array}}\end{array}$$

2. Coulomb Branch

2.1. **Recall the quiver case.** Let us consider the quiver gauge theory for two Q_0-graded vector spaces V, W

$$\text{gauge group } G = G_V = \prod_{i\in Q_0} GL(V_i),$$

$$\text{matters } \mathbf{N} = E_V \oplus L_{W,V} = \bigoplus_{i\to j} \mathrm{Hom}(V_i, V_j) \oplus \bigoplus_{i\in Q_0} \mathrm{Hom}(W_i, V_i),$$

$$\text{flavor group } F = G_W = \prod_{i\in Q_0} GL(W_i).$$

We can define two weights:

$$\text{highest weight } \lambda = \sum_{i\in Q_0} (\dim W_i)\Lambda_i, \qquad \Lambda_i = \text{fundamental weight}$$

$$\text{weight } \mu = \lambda - \sum_{i\in Q_0} (\dim V_i)\alpha_i, \quad \alpha_i = \text{simple root.}$$

By [BFN, Appendices], there is an algebra homomorphism

$$\begin{array}{c}\text{shifted Yangian of } \mathfrak{g}_Q \\ (\text{shifted by } \mu)\end{array} \longrightarrow \begin{array}{c}\text{the quantized} \\ \text{Coulomb branch}\end{array}$$

It was shown later in [Wee] that when the equivariant parameter $\hbar$ specialized to a nonzero complex number, this map is surjective.

In general, we can embed

$$\begin{array}{c}\text{quantized Coulomb} \\ \text{branch of } (G, \mathbf{N})\end{array} \longrightarrow \left(\begin{array}{c}\text{difference} \\ \text{algebra of } T\end{array}\right)^{W}_{\text{loc}}.$$

Different $\mathbf{N}$ have distinct "residue condition". The difference algebra is

$$\mathcal{D} = \bigoplus_{\lambda\in\Lambda} Q(\hbar) \cdot d_\lambda, \qquad \begin{array}{l} f(x,\hbar) d_\lambda \cdot g(x,\hbar) d_\mu \\ = f(x,\hbar) g(x + \lambda\hbar, \hbar) d_{\lambda+\mu}, \end{array}$$

where $\Lambda \subset \operatorname{Lie} T$ is the cocharacter lattice, and Q is the field of rational functions on $\operatorname{Lie} T$. We could view the difference algebra as the ring of operators on $Q(\hbar)$, with

$$\begin{aligned} f(x,\hbar) &= \text{left multiplication by } f(x,\hbar) \\ d_\lambda &= \text{difference operator } f(x,\hbar) \mapsto f(x+\hbar\lambda,\hbar). \end{aligned}$$

In the quiver case,

$$Q = \mathbb{C}(x_{i,j})_{i\in Q_0, 1\le j\le \dim V_i}, \qquad \mathcal{D} = \bigoplus_{i\in Q_0, 1\le j\le \dim V_i} Q(\hbar)\cdot d_{i,j}.$$

This in particular gives a GKLO type representation.

It worth mention that

$$\begin{aligned} E_i(u) &= \pm \text{monopole operator of } \varpi^\vee_{i,1} = \epsilon_{i,1} \\ F_i(u) &= \pm \text{monopole operator of } \varpi^\vee_{i,-1} = -\epsilon_{i,\mathbf{v}_i} \\ H_i(u) &\in \text{Gelfand–Tsetlin subalgebra} \\ &\quad \text{i.e. they do not contain difference operators in GKLO} \end{aligned}$$

2.2. **Quiver with involution.** Similar as above, we can consider

$$\begin{aligned} &\text{gauge group } G = G_V^\tau = \{(g_i)\in G_V : g_i^* = g_{\tau i}\} \\ &\text{matters } \mathbf{N} = E_V^\tau \oplus L_{W,V} = \operatorname{Alt}\left(\bigoplus_{i\to j} \operatorname{Hom}(V_i,V_j)\right) \oplus \bigoplus_{i\in Q_0} \operatorname{Hom}(W_i,V_i), \\ &\text{flavor group } F = G_W = \prod_{i\in Q_0} GL(W_i). \end{aligned}$$

Our result [SSX] is, there is an algebra homomorphism

$$\begin{array}{c}\text{twisted shifted} \\ \text{Yangian of } (\mathfrak{g}_Q, \mathfrak{g}_Q^\tau)\end{array} \longrightarrow \begin{array}{c}\text{the quantized} \\ \text{Coulomb branch}\end{array}.$$

The shift is a bit complicated:

$$\begin{aligned} &\sum_{j\in Q_0} (\dim W_j)(\Lambda_i + \Lambda_{\tau i}) - \sum_{i\to\tau i} (\Lambda_i + \Lambda_{\tau i}) - \sum_{j\in Q_0} (\dim V_j)\alpha_i \\ &= \lambda + \tau(\lambda) - \mu - \sum_{i\to\tau i} (\Lambda_i + \Lambda_{\tau i}). \end{aligned}$$

Note that the shift is spherical.

Remark. By [Wan], we can also use

$$\mathbf{N} = E_V^\tau \oplus L_{W,V} = \operatorname{Sym}\left(\bigoplus_{i\to j} \operatorname{Hom}(V_i,V_j)\right) \oplus \bigoplus_{i\in Q_0} \operatorname{Hom}(W_i,V_i),$$

The algebra homomorphism also exists, and the shift is

$$\sum_{j\in Q_0}(\dim W_j)(\Lambda_i+\Lambda_{\tau i})+\sum_{i\to\tau i}(\Lambda_i+\Lambda_{\tau i})-\sum_{j\in Q_0}(\dim V_j)\alpha_i$$
$$=\lambda+\tau(\lambda)-\mu+\sum_{i\to\tau i}(\Lambda_i+\Lambda_{\tau i}).$$

2.3. **Discussion of the proof.** To do computation, we have to make some choice. We pick an decomposition

$$Q_0=Q_0^-\sqcup Q_0^+,\qquad \text{with}\qquad \tau(Q_0^\pm)=Q_0^\mp.$$

Then we have

$$\begin{aligned}
G_V^\tau &\cong \prod_{i\in Q_0^+}GL(V_i), \qquad \text{the gauge group is of type A!}\\
E_V^\tau &\cong \bigoplus_{Q_0^+\ni i\to j\neq\tau(i)}\mathrm{Hom}(V_i,V_j)\oplus\bigoplus_{Q_0^+\ni i\to\tau i}\mathrm{Alt}(V_i,V_{\tau i}),\\
L_{W,V} &= \bigoplus_{i\in Q_0^+}\left(\mathrm{Hom}(W_i,V_i)\oplus\mathrm{Hom}(W_{\tau i},V_i^*)\right)
\end{aligned}$$

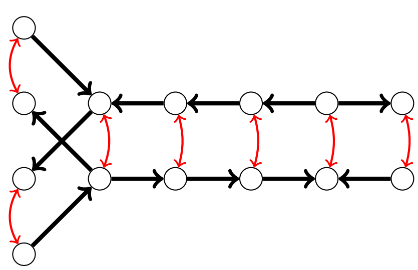
Under the algebra homomorphism

$$\begin{aligned}
&b_i(u)=\pm\text{monopole operator of }\varpi^\vee_{i,1}=\varpi^\vee_{\tau i,-1}\\
&h_i(u)\in\text{Gelfand–Tsetlin subalgebra}\\
&\qquad\text{i.e. they do not contain difference operators in GKLO}
\end{aligned}$$

The formula of $b_i(u)$ is

$$\sum_{r=1}^{v_i}\underbrace{\frac{1}{-u-x_{i,r}-\frac{\hbar}{2}}}_{\text{coefficients}}\underbrace{\prod_{i\to\tau i}V_{\tau i,r}(x_{i,r}+\tfrac{\hbar}{2})}_{\mathrm{Alt}(V_i,V_{\tau i})}\underbrace{\prod_{i\to j\neq\tau(i)}V_j(x_{i,r}+\tfrac{\hbar}{2})}_{\mathrm{Hom}(V_i,V_j)}\underbrace{W_{\tau i}(-x_{i,r}-\tfrac{\hbar}{2})}_{\mathrm{Hom}(V_i,W_{\tau i})}\underbrace{\frac{1}{V_{i,r}(x_{i,r})}d_{i,r}}_{\text{affine Grassmannian}}$$

Example (Diagonal Type). Assume that the quiver Q is the disjoint union of two identical copies, denoted Q^+ and Q^-. Let $\tau: Q\to Q$ be the involution that interchanges the vertices in corresponding positions.



Let $V^+ = \bigoplus_{i \in Q_0^+} V_i$ be the Q_0^+-graded vector space. Then we have isomorphisms

$$G_V^\tau \simeq G_{V^+}, \qquad E_V^\tau \simeq E_{V^+}.$$

In this case, our result recover the result for quiver guage theory.

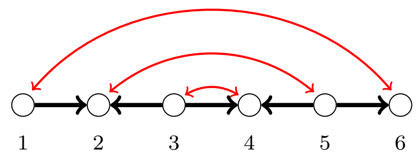
Example. Consider a quiver with involution obtained from a Satake diagram of type AIII

$$Q_0 = \{1, \dots, 2n\}, \qquad \tau i = 2n + 1 - i$$

and Q_1 is chosen such that

$$\#\big\{h \in Q_1 : \{s(h), t(h)\} = \{i, j\}\big\} = \delta_{|i-j|,1}.$$

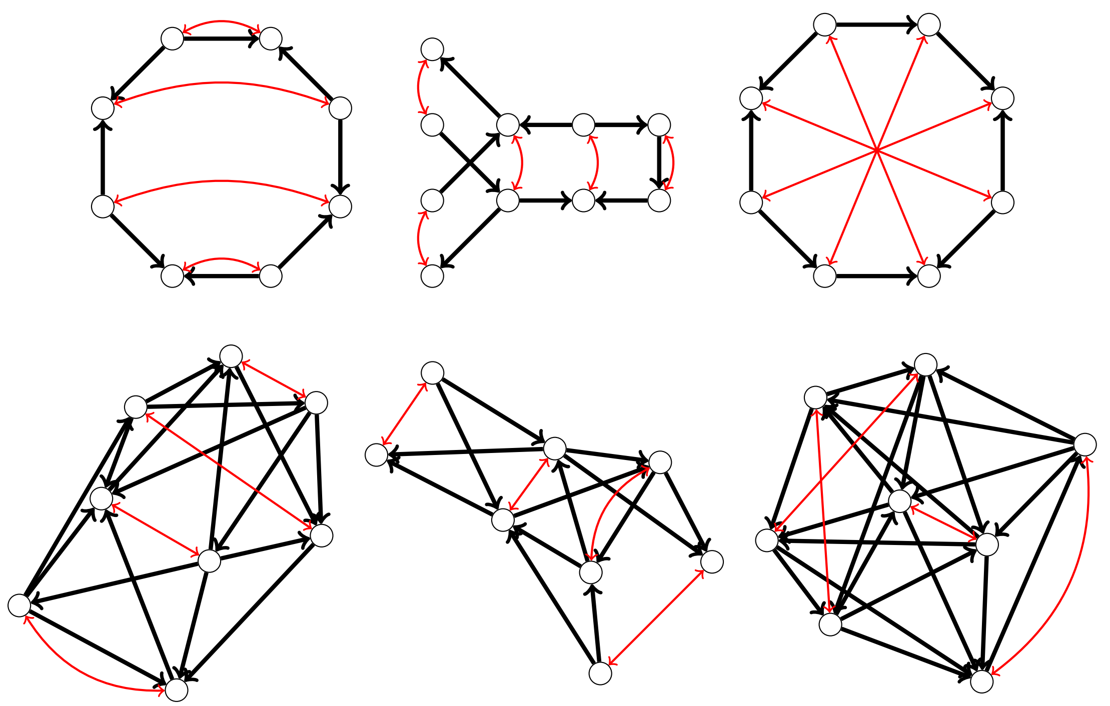
We can choose $Q_0^+ = \{1, \dots, n\}$ and $Q_1^+ = \{h \in Q_1 : \max\{s(h), t(h)\} \leq n\}$. For example, when $n = 3$ we have



Then we have

$$\begin{aligned} G_V^\tau &\cong GL(v_1) \times GL(v_2) \times GL(v_3) \\ E_V^\tau &\cong \mathrm{Hom}(\mathbb{C}^{v_1}, \mathbb{C}^{v_2}) \oplus \mathrm{Hom}(\mathbb{C}^{v_3}, \mathbb{C}^{v_2}) \oplus \Lambda^2(\mathbb{C}^{v_3}). \end{aligned}$$

Example. We remark that the quiver can be very general, for example, it can be of the following types:



2.4. **Finishing the proof.** We need some guess for the expression of $h_i(u)$. This is essentially by experimental computation by computer. The relations are tediously examined one-by-one.

- Most of relations are proved by combinatorial identities.
- The most complicated relation is the $\imath$Serre relation, where we used a trick.

Combinatorial identity. For a Laurant series $f(z) = \sum_{i\in\mathbb{Z}} a_i z^i$ in z^{-1} , we denote the truncation

$$\big(f(z)\big)^\circ := \sum_{i<0} a_i z^i.$$

Assume $f(z) = \dfrac{g(z)}{\prod_{i=1}^n (z-z_i)}$, where $z_1, \ldots, z_n$ are distinct and $g(z) \in \mathbb{C}[z]$. Then

$$(f(z))^\circ = \sum_{i=1}^n \frac{1}{z-z_i} \frac{g(z_i)}{\prod_{j\neq i}(z_i - z_j)}.$$

Exericse. Prove it.

Trick on Serre relations. Different from the case of shifted Yangian, we need to formulate a shifted version of $\imath$Serre relation. This relation cannot be formulated in terms of generating functions (the one in [LZ] is not correct). Actually, we know how to do this from the computation, and it turns out the shifted version is

$$\mathrm{Sym}_{k_1,k_2}\left[b_{i,k_1}, [b_{i,k_2}, b_{\tau i,r}]\right] = \frac{4}{3}\,\mathrm{Sym}_{k_1,k_2} (-1)^{k_1} \sum_{p=0}^\infty 3^{-p}\left[b_{i,k_2+p}, h_{\tau i,k_1+r-p}\right].$$

However, it is still very hard to prove it.

In the Yangian case, there is a classical principle that

$$\text{affine Serre relation} \xLeftarrow{\text{other relations}} \text{finite Serre relation}.$$

By [LZ], we can make a similar one for $\imath$Serre relations. After generalizing this principle to the shifted version, we reduce to check

$$\hbar^{-1}[b_{i,0}, [b_{i,0}, b_{\tau i,0}]] = \frac{4}{3}\sum_{p=0}^\infty 3^{-p}[b_{i,p}, h_{\tau i,-p}],$$

or

$$\hbar^2[b_{i,0}, [b_{i,0}, b_{\tau(i)}(v)]] = \big(4v\hbar[b_i(3v), h_{\tau(i)}(v)]\big)^\circ.$$

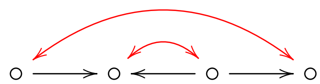
Now it becomes possible to check.

2.5. **Philosophical Remark.** It is typical that in 3D mirror symmetry that two mirror symmetric objects are defined by the same combinatorial object, while no direct relation between them.

In our case, associated with a quiver with involution, on one hand, we have a module of Enomoto–Kashiwara algebra, and on the other hand, we have a truncated shifted twisted Yangian.

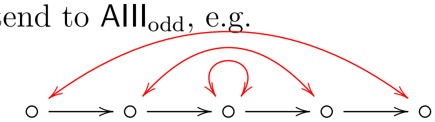
3. General Quasi-split Types

Among all quasi-split Satake diagrams of finite type, only $\mathsf{AIII}_{\text{even}}$ has no fixed points. For example,

AIII_4 

It is natural to ask what is the replacement for those Satake diagram with fixed points.

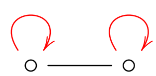
3.1. **Naïve way.** We can relax the definition of quivers with involution by allowing fixed vertices. This can be used to extend to $\mathsf{AIII}_{\text{odd}}$, e.g.

AIII_3 

For graded vector spaces V and W, we still need to identify $V_i^* \cong V_{\tau(i)}^*$. But for fixed i, we need to determine it is by orthogonal or symplectic form, and the corresponding contribution of group G will be orthogonal or symplectic. For example, in the above example, we will have

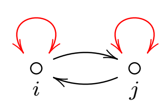
$$\begin{aligned} G_V^\tau &= GL(v_1) \times GL(v_2) \times O/Sp(v_3) \\ E_V^\tau &= \mathrm{Hom}(\mathbb{C}^{v_1}, \mathbb{C}^{v_2}) \oplus \mathrm{Hom}(\mathbb{C}^{v_2}, \mathbb{C}^{v_3}). \end{aligned}$$

But it cannot realize other finite types such as

AI_2 

Notice that if $i \neq j$ are both fixed, then

$$\#\{i \to j\} = \#\{j \to i\}$$



since we require the involution reverse the orientation.

3.2. **Nakajima's suggestion.** In this section, we mainly follow the suggestion of [Nak, 3(ix)]. There are three cases

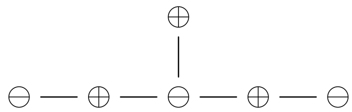
(C) Second exterior/symmetric power at the end of the quiver.
(A) Orthosymplectic quivers.
(B) Mixture of orthosymplectic and usual quivers.

They correspond to

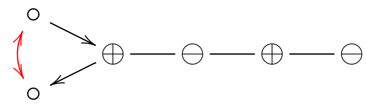
(C) No-fixed-vertex case — the diagram involution has no fixed vertices;
(A) Split case — the diagram involution is trivial;
(B) Other cases.

The suggestion in case (C) is exactly what we discussed today. We will discuss the case (A) orthosymplectic quiver. The case (B) is essentially (A) mixed with the naïve extension above.

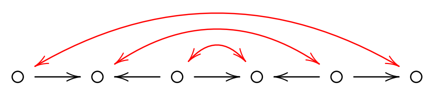
The following are examples in finite types:



(A) Orthosymplectic quivers



(B) Mixture of (A) and (C)



(C) Quiver with involution

3.3. **Everything symplectic.** We want to point out the Higgs branch and Coulomb branch are both defined for symplectic representations $(G, \mathbf{M})$, i.e.

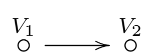
$$\text{group homomorphism } G \to Sp(\mathbf{M}).$$

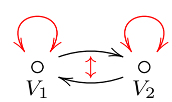
The current case $(G, \mathbf{N})$ is known as the *cotangent type*, and

$$\mathbf{M} = T^*\mathbf{N} = \mathbf{N} \oplus \mathbf{N}^*, \qquad G \to GL(\mathbf{N}) \subset Sp(\mathbf{M}).$$

- The Higgs branch is defined by the Hamitonian reduction.
- The Coulomb branch of non-cotangent type is constructed, but in a very complicated way.

3.4. **Involution on M — an example.** Let us consider





The quiver gauge theory of the LHS is

$$\begin{aligned} G &= GL(V_1) \times GL(V_2), \\ \mathbf{N} &= \mathrm{Hom}(V_1, V_2), \\ \mathbf{M} &= \mathrm{Hom}(V_1, V_2) \oplus \mathrm{Hom}(V_2, V_1). \end{aligned}$$

The trick is the $\mathbf{M}$ can be viewed as the $\mathbf{N}$ for the RHS — the doubled quiver. We have an involution

$$\tau : (f, g) \longmapsto \pm(g^*, f^*).$$

Note that

V_1, V_2 are both O/Sp $\Rightarrow \tau$ is not symplectic	$V_1 \& V_2$ are O&Sp or Sp&O $\Rightarrow \tau$ is not symplectic

In the second case, we have

$$G^\tau = O(V_1) \times Sp(V_2) \text{ or } Sp(V_1) \times O(v_1)$$
$$\mathbf{M}^\tau \cong V_1 \otimes V_2.$$

This is the typical source of non-cotangent type.

Linear Algebra Exercise. If V_1 is an O-space and V_2 is a Sp-space, then $V_1 \otimes V_2$ is a Sp space with

$$\omega(v_1 \otimes v_2, v_1' \otimes v_2') = \langle v_1, v_1' \rangle \cdot \omega(v_2, v_2')$$

and $O(V_1) \times Sp(V_2) \subset Sp(V_1 \otimes V_2)$.

3.5. **Orthosymplectic quivers.** The name is confusing, actually there is no arrow in an OSp quiver. Instead, we bipartite the Dynkin diagram by

$$Q_0 = \{\text{orthogonal vertices}\} \sqcup \{\text{symplectic vertices}\}.$$

Let V, W be two Q_0-graded vector spaces such that

$$\begin{matrix} V_i \text{ is equipped with O/Sp form,} \\ W_i \text{ is equipped with Sp/O form,} \end{matrix} \quad \text{if the vertex } i \text{ is O/Sp.}$$

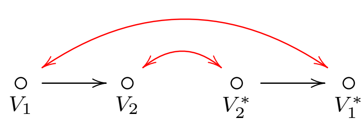
Then

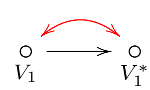
$$\text{gauge group } G_V = \prod_{i \text{ is O}} O(V_i) \times \prod_{i \text{ is Sp}} Sp(V_i),$$
$$\text{flavor group } G_W = \prod_{i \text{ is O}} Sp(W_i) \times \prod_{i \text{ is Sp}} O(W_i),$$
$$\text{matters } \mathbf{M} = \bigoplus_{i - j} V_i \otimes V_j \oplus \bigoplus_{i \in Q_0} V_i \otimes W_i.$$

More information can be found in Nakajima's talk [link].

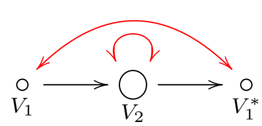
3.6. **Summary.** The contribution can be summarized in the following diagrams

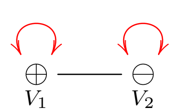
$$G = GL(V_1) \times GL(V_2) \qquad G = GL(V_1)$$
$$\mathbf{N} = \mathrm{Hom}(V_1, V_2) \qquad \mathbf{N} = \Lambda^2 V_1 \text{ or } S^2 V_1$$





$$G = GL(V_1) \times O/Sp(V_2) \qquad G = O(V_1) \times Sp(V_2)$$
$$\mathbf{N} = \mathrm{Hom}(V_1, V_2) \qquad \mathbf{M} = V_1 \otimes V_2$$





3.7. **Expectation.**

Coulomb branch. In [LWW], it is expected that

- the GKLO representation loc.cit. agrees with the Coulomb branch of $(G, \mathbf{M})$ introduced above.

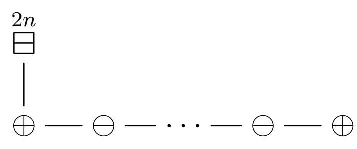
Our result partially justifies a special case when the involution has no fixed vertices.

Higgs branch. In [Nak, Li], it is proved that

- The twisted Yangian acts on the equivariant cohomology of Higgs branch of $(G, \mathbf{M})$.

However, the twisted Yangian is NOT the corresponding Satake type. The quiver involution is distinct by $-w_0$. We also expect that the Higgs branch should also be related to EK algebra.

Example. It is known that the cotangent bundle of type C partial flag varieties corresponds to the Higgs branch of the following OSp quiver



The quiver is of type AI, while by [SY], the cohomology of type C Springer resolution admits an action of twisted Yangian of type AIII.

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